

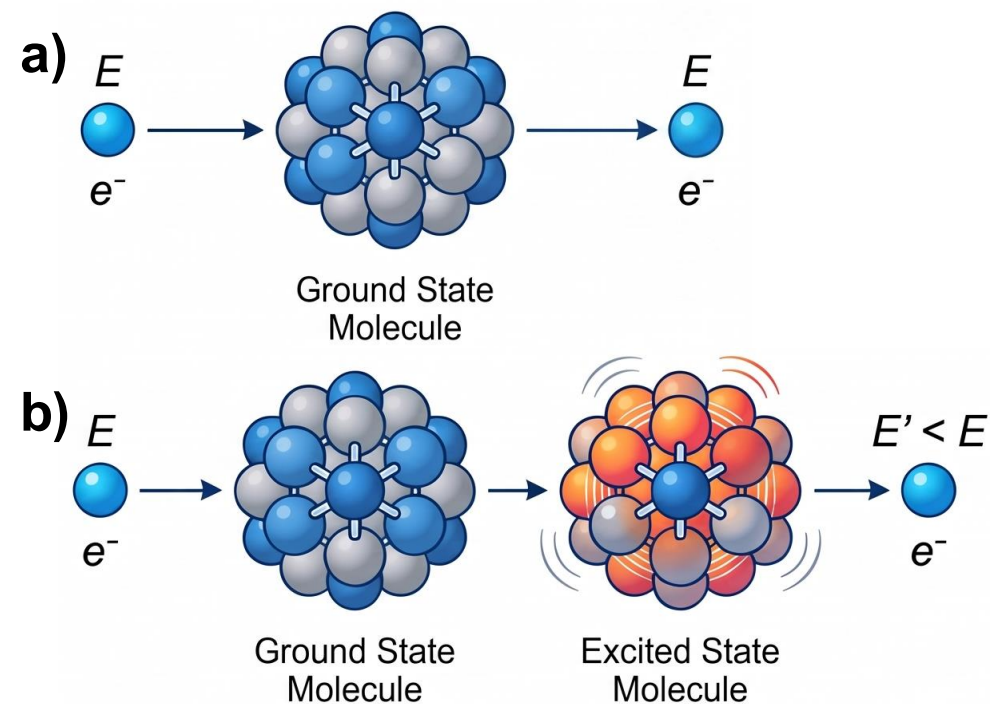
Ab Initio Electron-Molecule Scattering Calculations for Methane Using UKRMol+ and ASTRA

Rachel Lee, Dr. Siddhartha Chattopadhyay, and
Dr. Thomas C. Allison

Material Measurement Laboratory / Material Data Division /
Theory, Modeling, and Simulation Group

Scattering calculations are needed to model processes

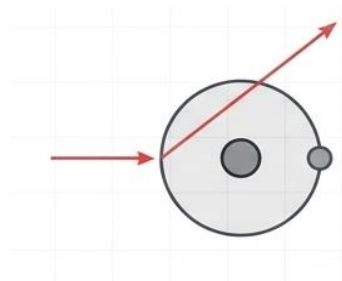
- Electron-molecule collisions in plasma processing, semiconductor manufacturing, and atmospheric chemistry
- **Cross sections** tell us the total probabilistic target area for interaction
 - **Integral Cross Section:** the effective target area for one specific channel (elastic scattering)
 - **Total Cross Section:** sum of probabilities of all channels (elastic + all inelastic)



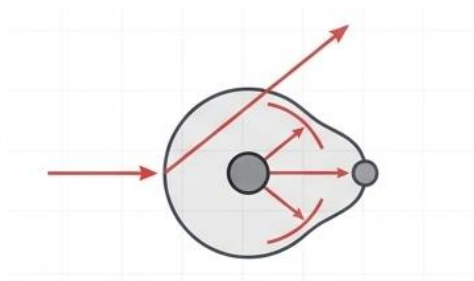
(a) Schematic of elastic scattering: $e^- + \text{CH}_4 \rightarrow e^- + \text{CH}_4$. Scattering electron's kinetic energy and internal molecular state remain unchanged.

(b) Schematic of inelastic scattering: $e^- + \text{CH}_4 \rightarrow e^- + \text{CH}_4^*$, where $*$ denotes the excited state. Net change in energy to the target molecule.

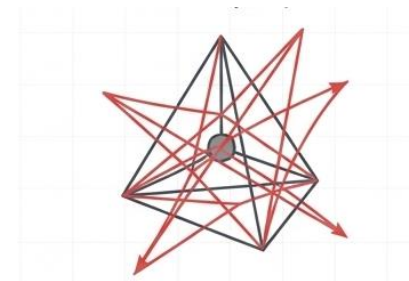
Computational Methods



**Static
Exchange (SE)**



**Static Exchange +
Polarization (SEP)**

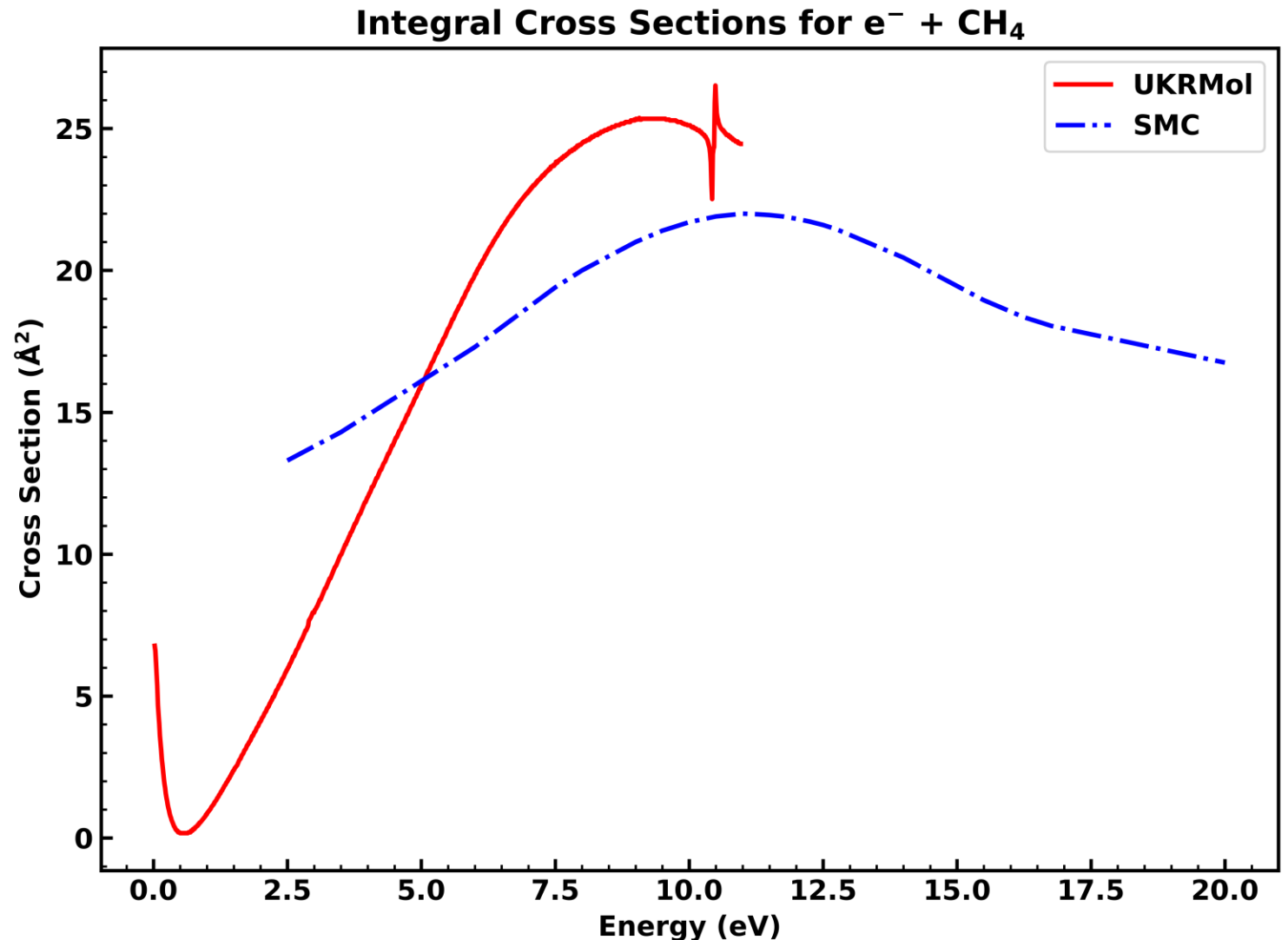


**Complete Active Space (CAS)
(Close-Coupling, R-Matrix)**

	Static Exchange (SE)	Static Exchange + Polarization (SEP)	Complete Active Space (CAS) (Close-Coupling, R-Matrix)
Physical Assumption	Target is frozen	Target is induced dipole	Complex correlation
Interactions Captured	Ground-state potential scattering	Long-range attractive force and some target distortion	Multi-reference configuration interaction. Models internal electronic shifts, captures resonances, and enables impact dissociation
Computational Cost	Lowest	Medium	Highest

SE vs CAS Calculation

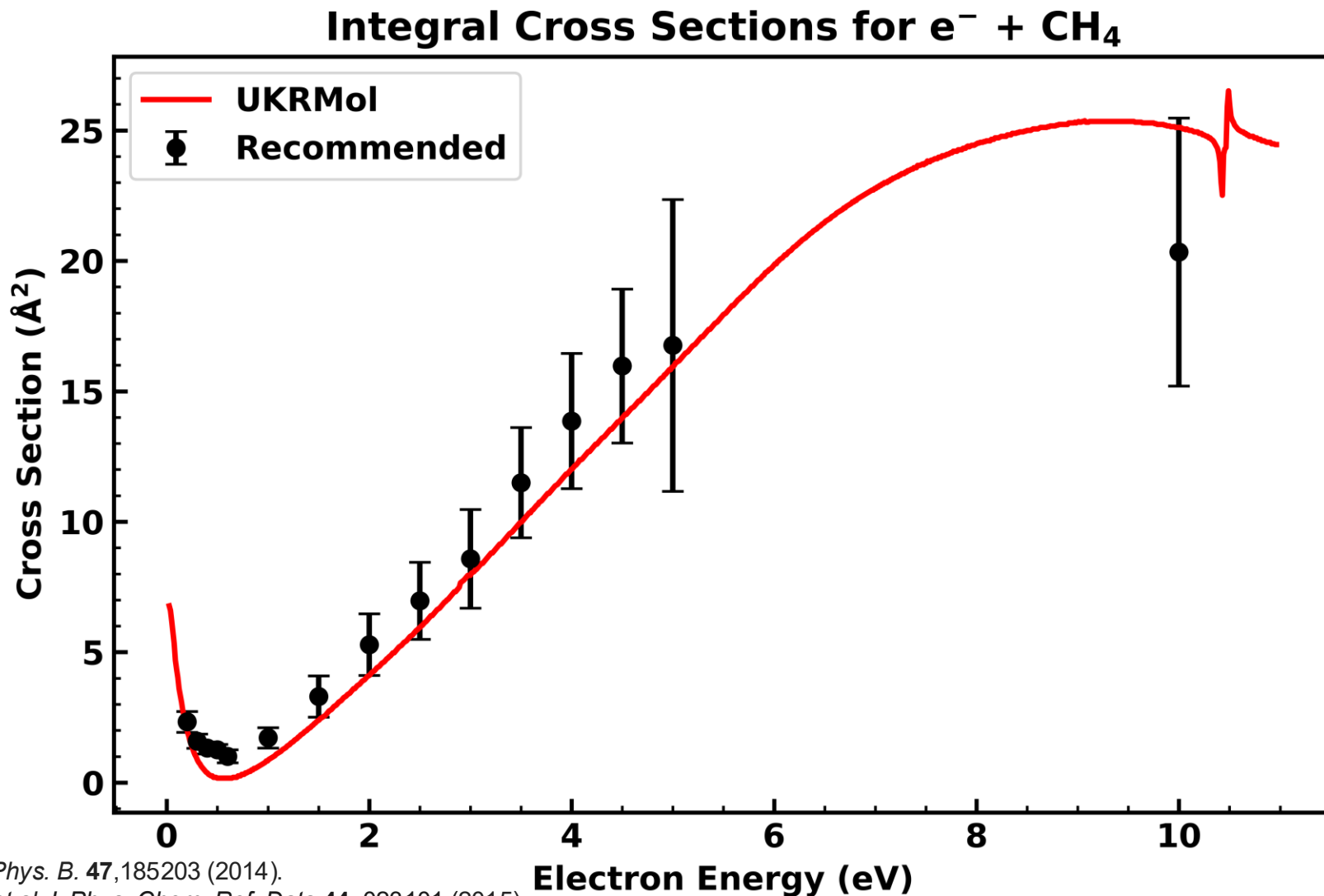
Both curves show calculated cross sections for the elastic scattering of methane



[UKRMol]: Brigg *et al J. Phys. B.* **47**, 185203 (2014).

[SMC]: Lima *et al Phys. Rev. A.* **32**, 2696–2701 (1985).

What do we look for in a cross section plot?

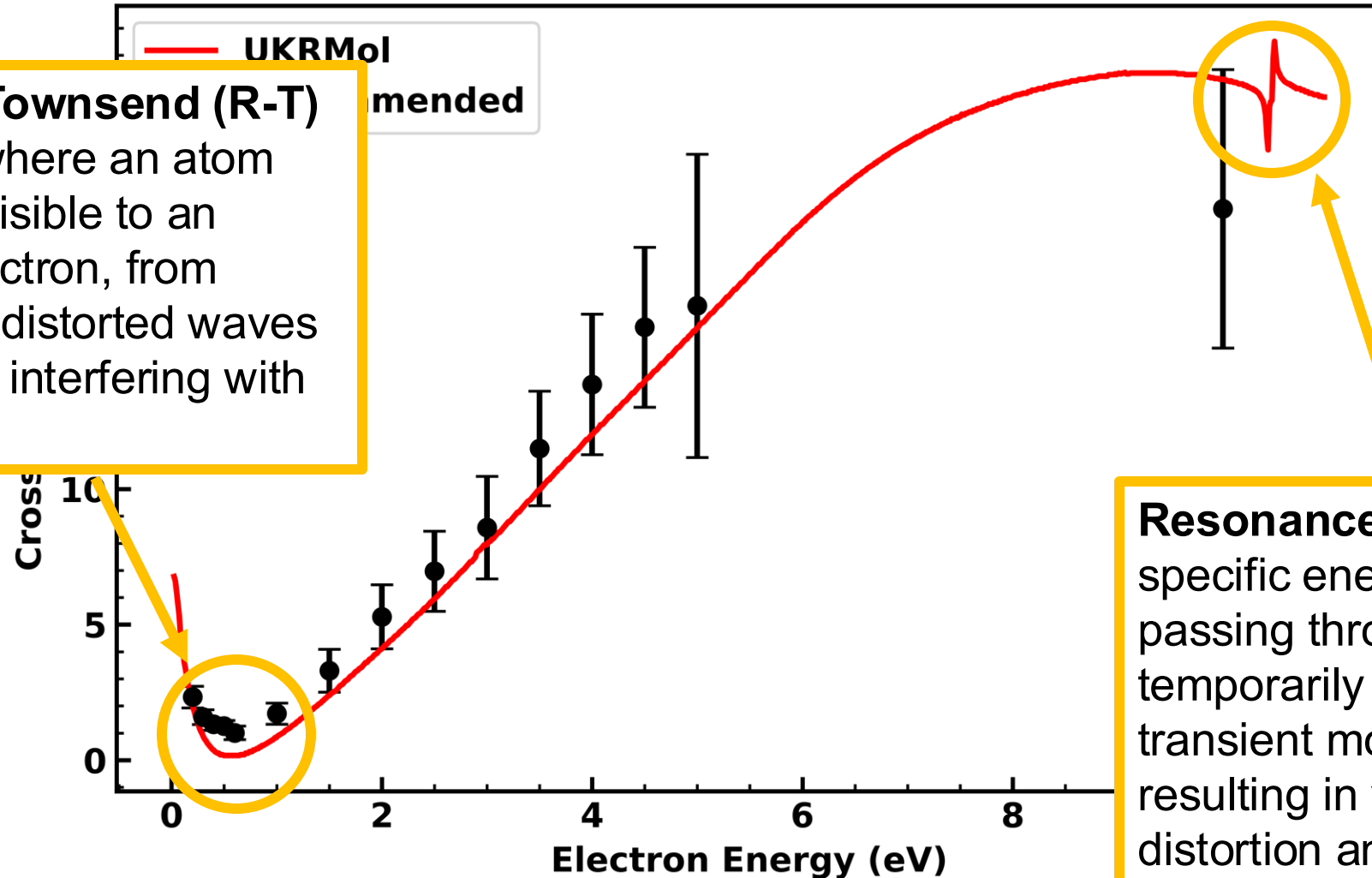


[UKRMol]: Brigg *et al* *J. Phys. B.* **47**,185203 (2014).

[Recommended]: Song *et al* *J. Phys. Chem. Ref. Data* **44**, 023101 (2015).

What do we look for in a cross section plot?

Integral Cross Sections for $e^- + \text{CH}_4$

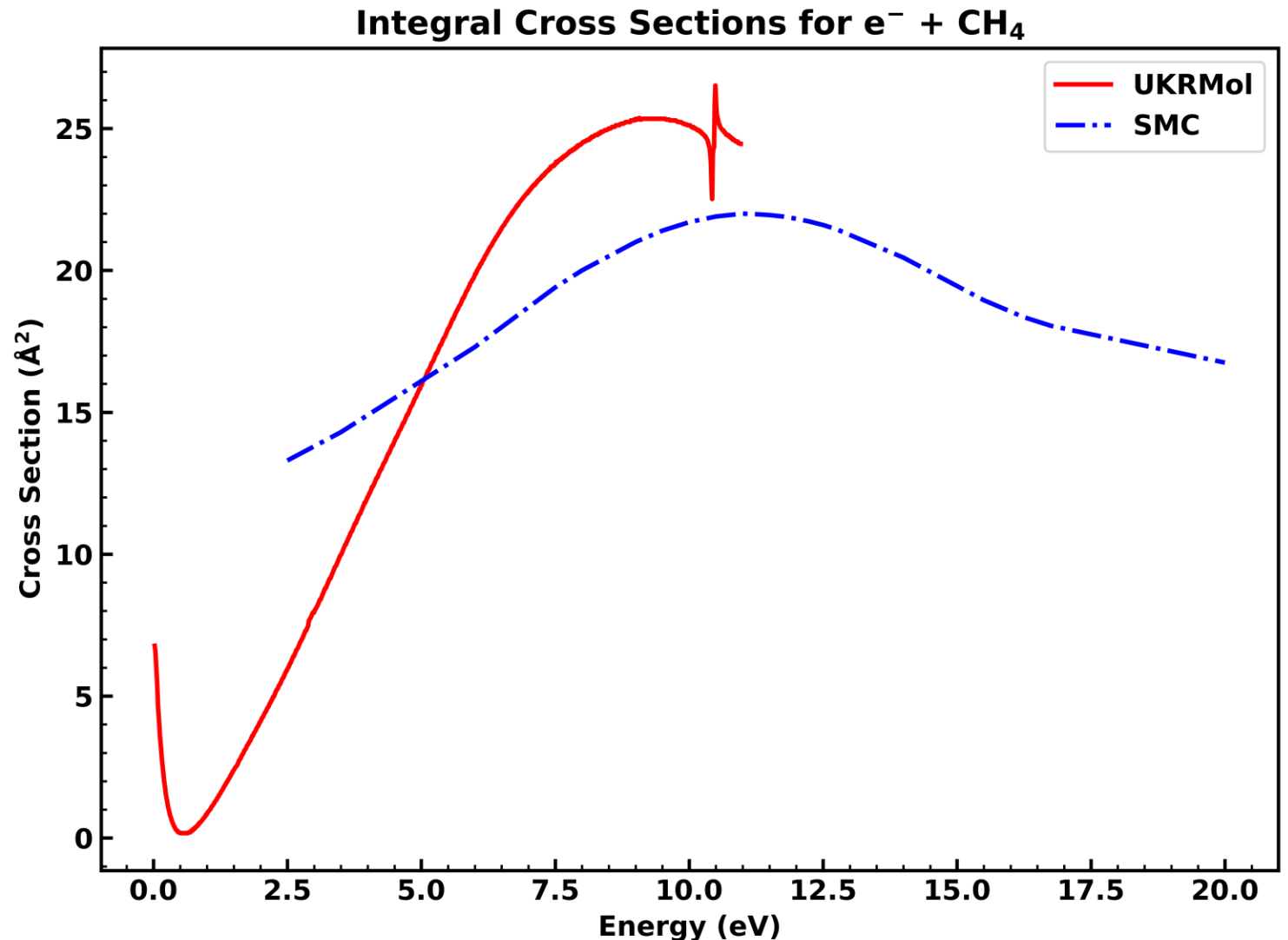


Ramsauer-Townsend (R-T) minimum: where an atom becomes invisible to an incoming electron, from incident and distorted waves destructively interfering with each other

Resonance peaks: at specific energies, rather than passing through electrons are temporarily captured in transient molecule orbitals, resulting in vibrational distortion and dissociation

SE vs CAS Calculation

Both curves show calculated cross sections for the elastic scattering of methane



[UKRMol]: Brigg *et al J. Phys. B.* **47**, 185203 (2014).

[SMC]: Lima *et al Phys. Rev. A.* **32**, 2696–2701 (1985).

Methods for calculating cross sections

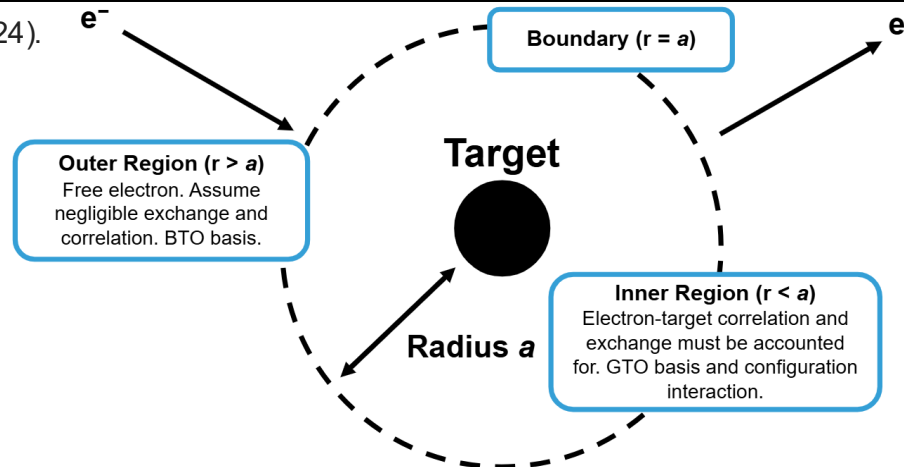
UKRMol+ (well-established)

- **R-Matrix Theory:** Divides space into an inner region (correlation and exchange effects) and an outer region (multipole expansion)
- Target states computed using **HF, CASSCF, or MASSCF**
- Target wavefunctions represented using **CASCI**

ASTRA (new)

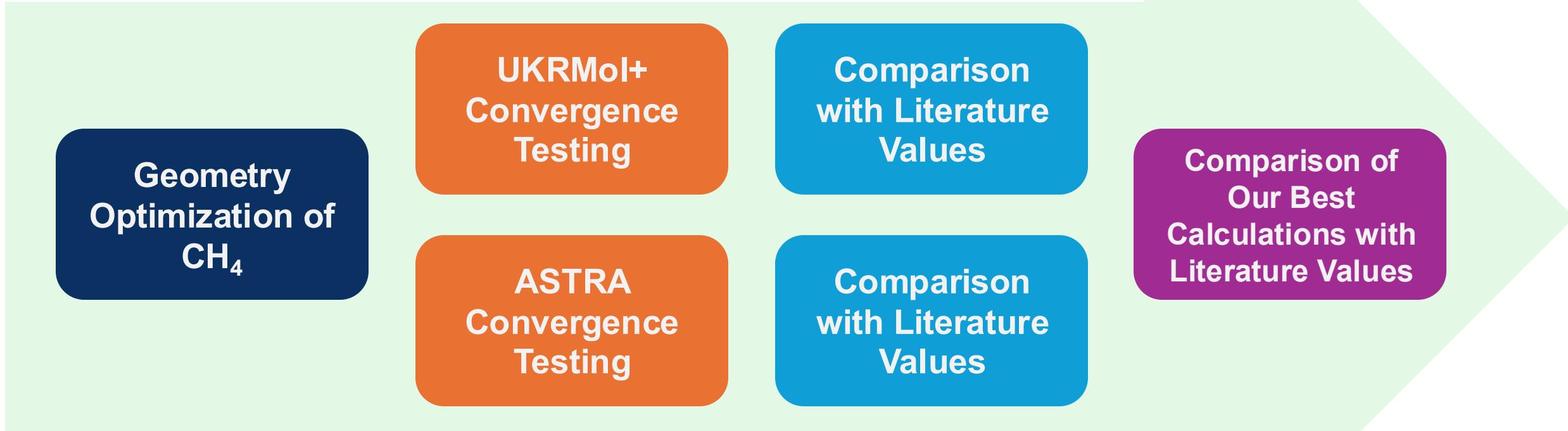
- **TDMCC Formalism:** Coupling computed via many-body transition-density matrices
- Lippmann-Schwinger equation solved via direct continuum-basis expansion of scattering wavefunction
- Target states optimized through **state-averaged (SA) MCSCF**, represented using **CASCI**
- Exchange terms treated exactly; computational cost is independent of CI space

Houfek et al *Comput. Phys. Commun.* **298**, 109113 (2024).

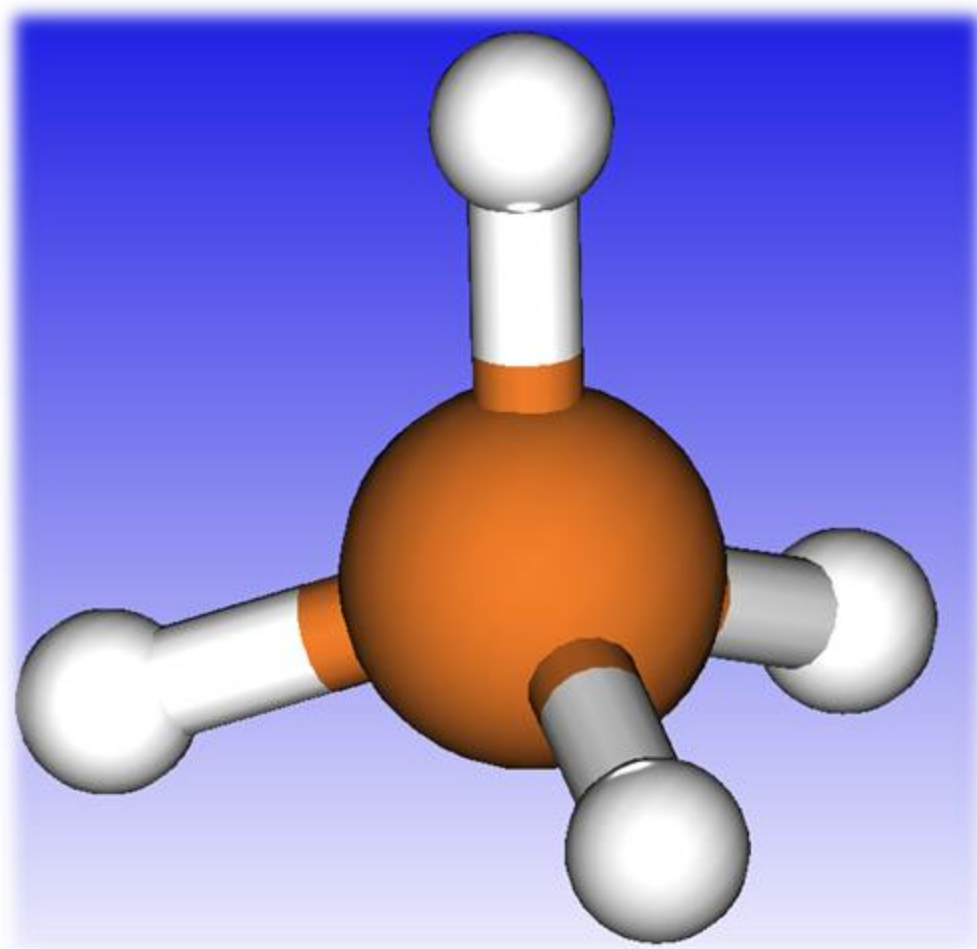


Randazzo et al *Phys. Rev. Research* **5**, 043115 (2023).

Overview of Results



Geometry Optimization



SCF geometry optimization calculations

- cc-pVTZ basis set
- D_2 and C_{2v} symmetries

Software	Symmetry Group	Energy (a.u.)
ASTRA	C_{2v}	-40.2133
UKRMol+	C_{2v}	-40.2133
UKRMol+	D_2	-40.2135

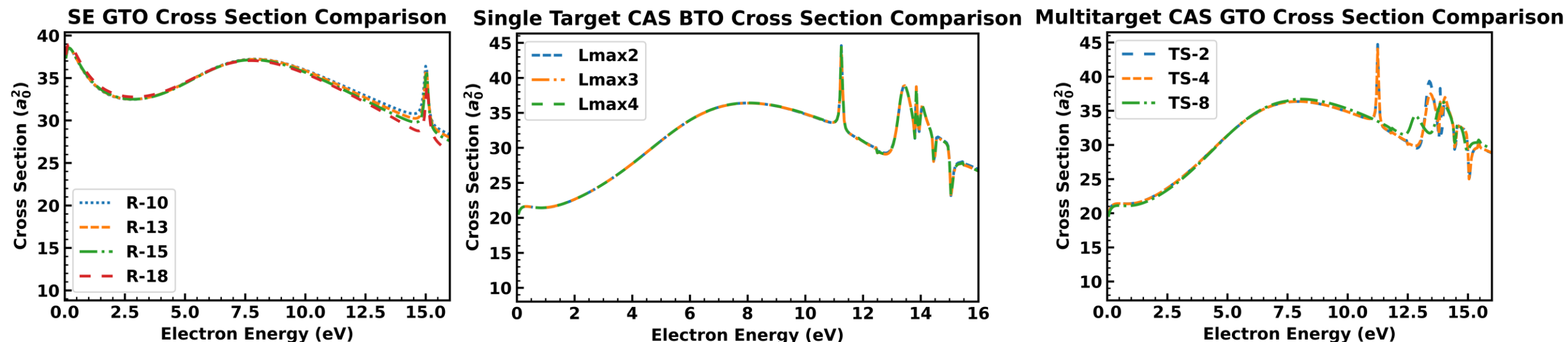
Benchmarking Calculated Energies

Ground state energies for CH₄ with C_{2v} symmetry reported in the NIST Computational Chemistry Comparison and Benchmark DataBase:

- CID/cc-pVTZ: -40.421 a.u.
- CID/aug-cc-pVQZ: -40.434 a.u.

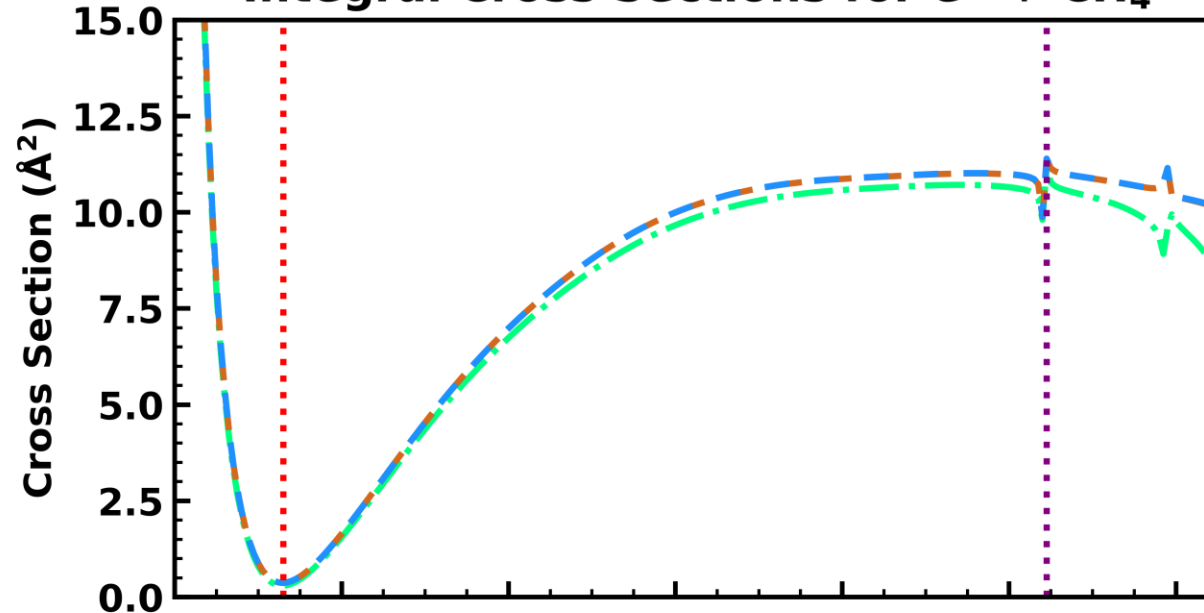
Target States	UKRMol+-CAS(8,8) (a.u.)	$\Delta E_{\text{UKRMol+}}$ (eV)	ASTRA-CAS(8,8) (a.u.)	ΔE_{ASTRA} (eV)
¹ A ₁	-40.220	0.000	-40.249	0.000
³ A ₁ / ³ B ₁ / ³ B ₂	-39.803	11.347	-39.890	9.769
¹ A ₁ / ¹ B ₁ / ¹ B ₂	-39.781	11.946	-39.878	10.095
³ A ₁	-39.755	12.653	-39.796	12.327

UKRMol+ Convergence Testing



Close agreement found with BTO and GTO calculations and calculations using different GTO radii and BTO Lmax values (convergence up to 10 eV)

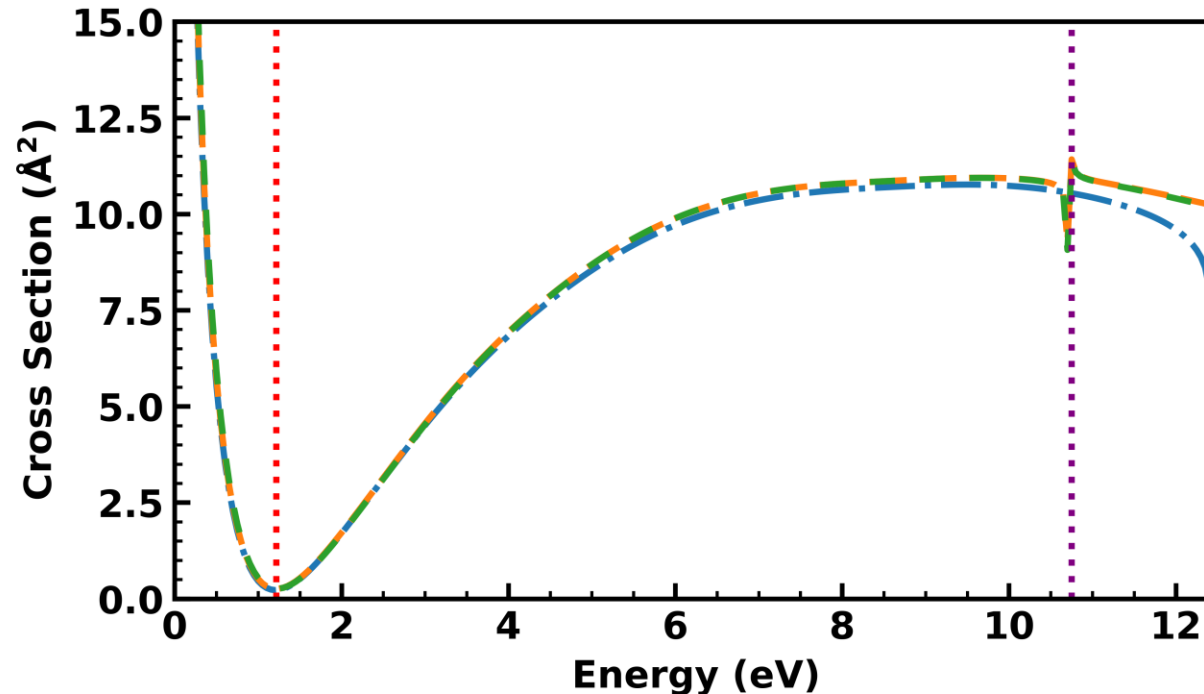
Integral Cross Sections for $e^- + \text{CH}_4$



Literature Values:

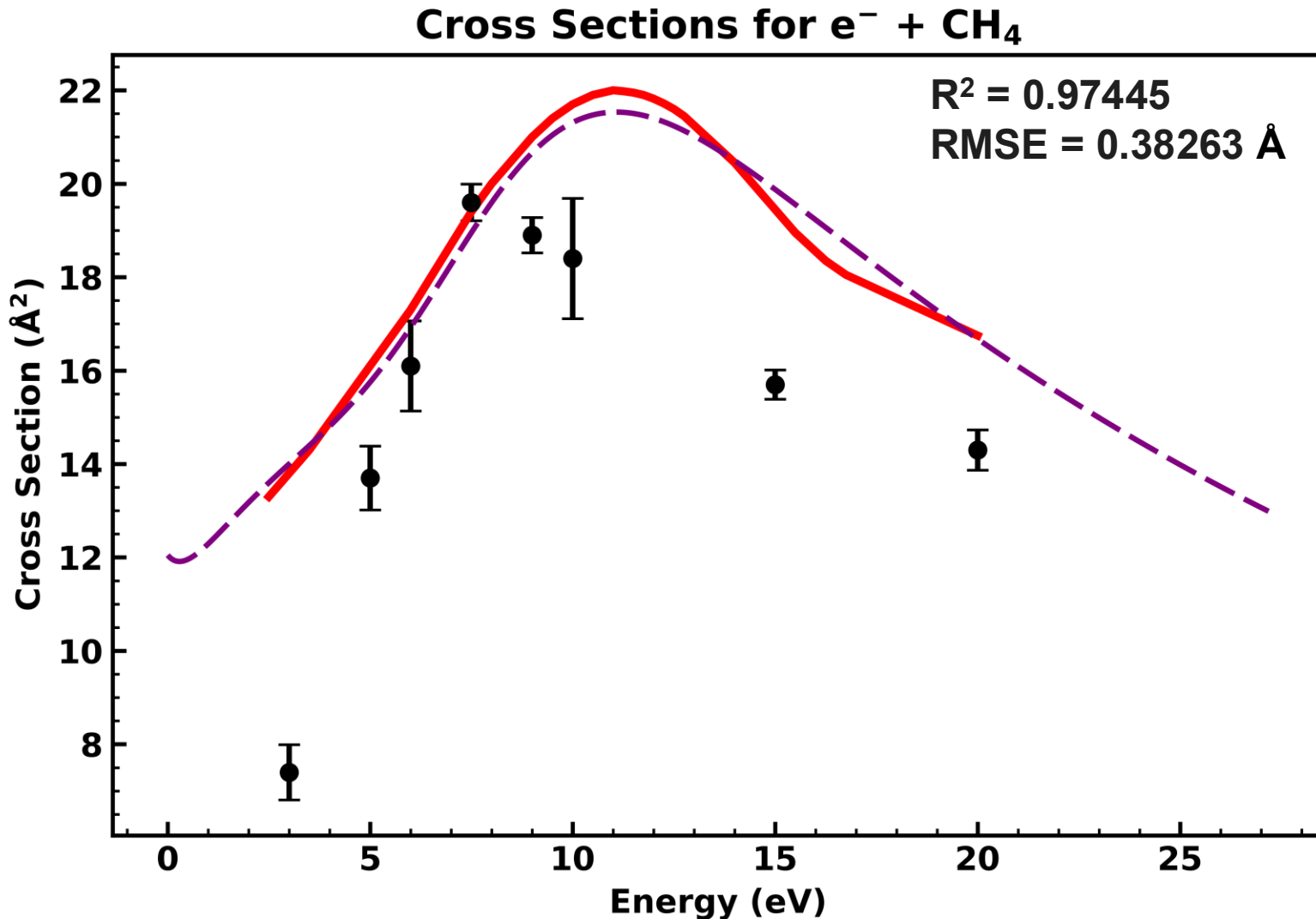
- R-T minimum at 0.4 eV
- Resonance peak at 10.2 eV

- UKRMol+-CAS(8,8)-TS8-Lmax2
- UKRMol+-CAS(8,8)-TS8-Lmax4
- UKRMol+-CAS(8,8)-TS8-Lmax6
- Average minimum: 1.301 eV
- Average peak: 10.451 eV



- UKRMol+-CAS(8,8)-TS1-Lmax6
- UKRMol+-CAS(8,8)-TS2-Lmax6
- UKRMol+-CAS(8,8)-TS4-Lmax6
- Average minimum: 1.218 eV
- Average peak: 10.751 eV

ASTRA SE Validation



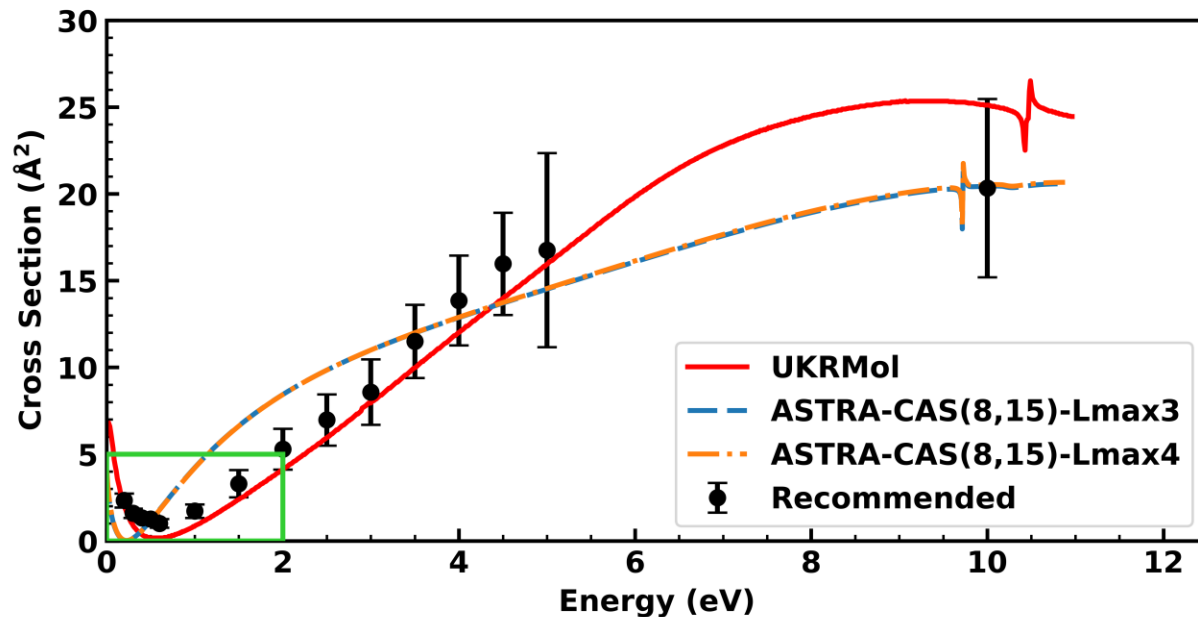
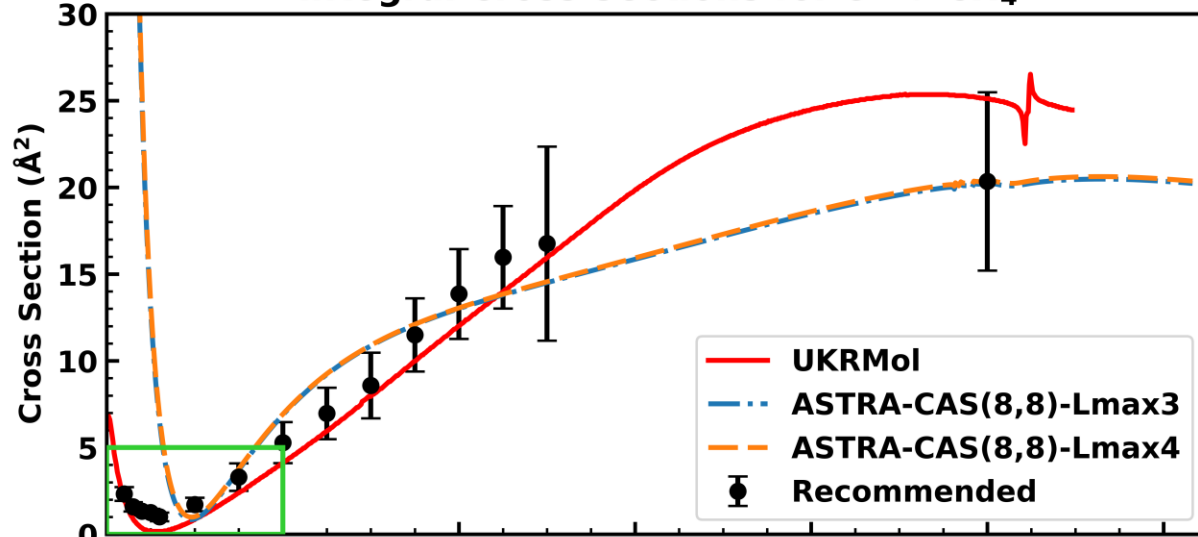
Calculation Parameters:

- cc-pVTZ basis set
- $L_{\text{max}} = 3$
- Single target state
- Ground state energy = -40.213 a.u.

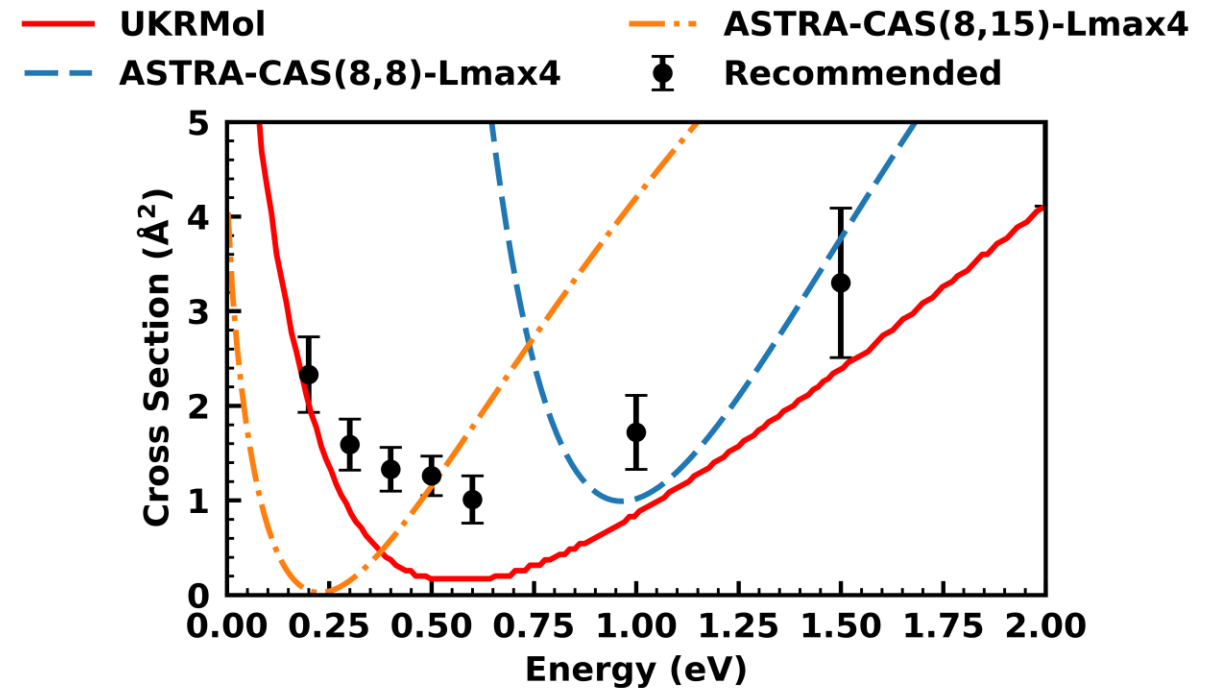
[SMC]: Lima *et al* *Phys. Rev. A* **32**, 2696–2701 (1985).
[CBES]: Tanaka *et al* *J. Phys. B* **15**, 3305–3319 (1982).

ASTRA Convergence Testing

Integral Cross Sections for $e^- + \text{CH}_4$



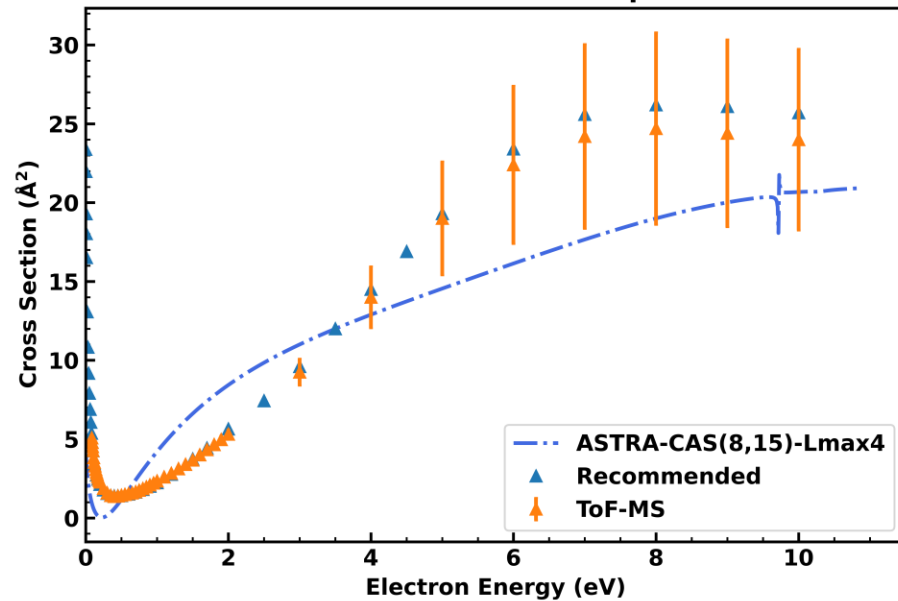
Integral Cross Sections for $e^- + \text{CH}_4$



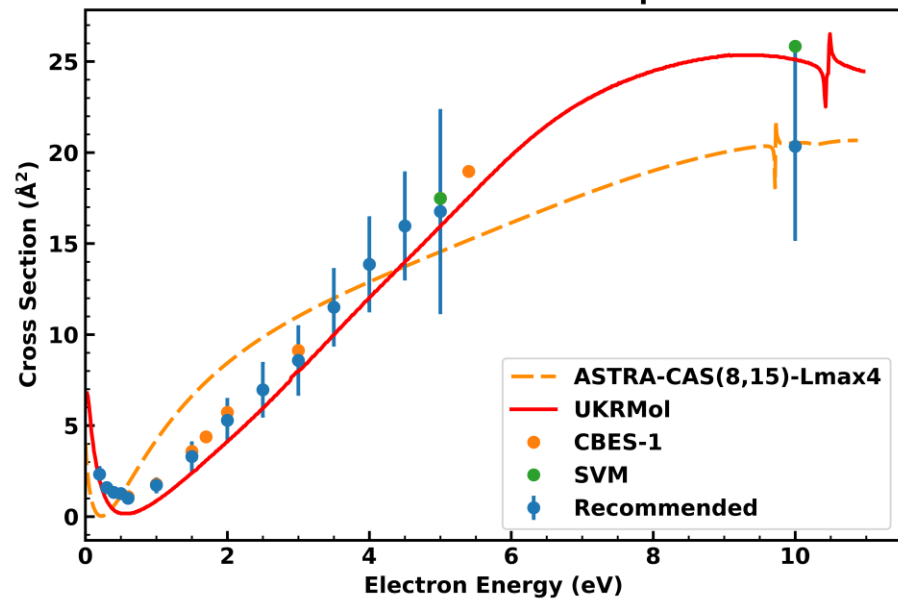
[UKRMol]: Brigg *et al* *J. Phys. B.* **47**,185203 (2014).

[Recommended]: Song *et al* *J. Phys. Chem. Ref. Data* **44**, 023101 (2015).

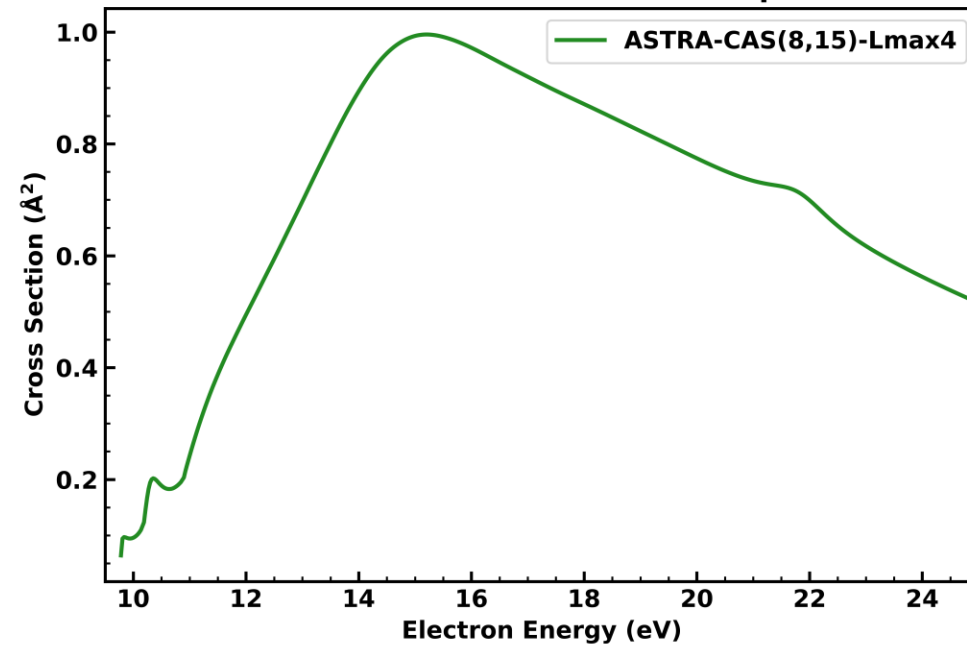
Total Cross Sections Comparison



Elastic Cross Sections Comparison



Total Inelastic Cross Sections Comparison



[CBES-1]: Bundschu *et al* *J. Phys. B.* **30**, 2239 (1997).

[Recommended]: Song *et al* *J. Phys. Chem. Ref. Data* **44**, 023101 (2015).

[SVM]: Cho *et al* *J. Phys. B.* **41**, 045203 (2008).

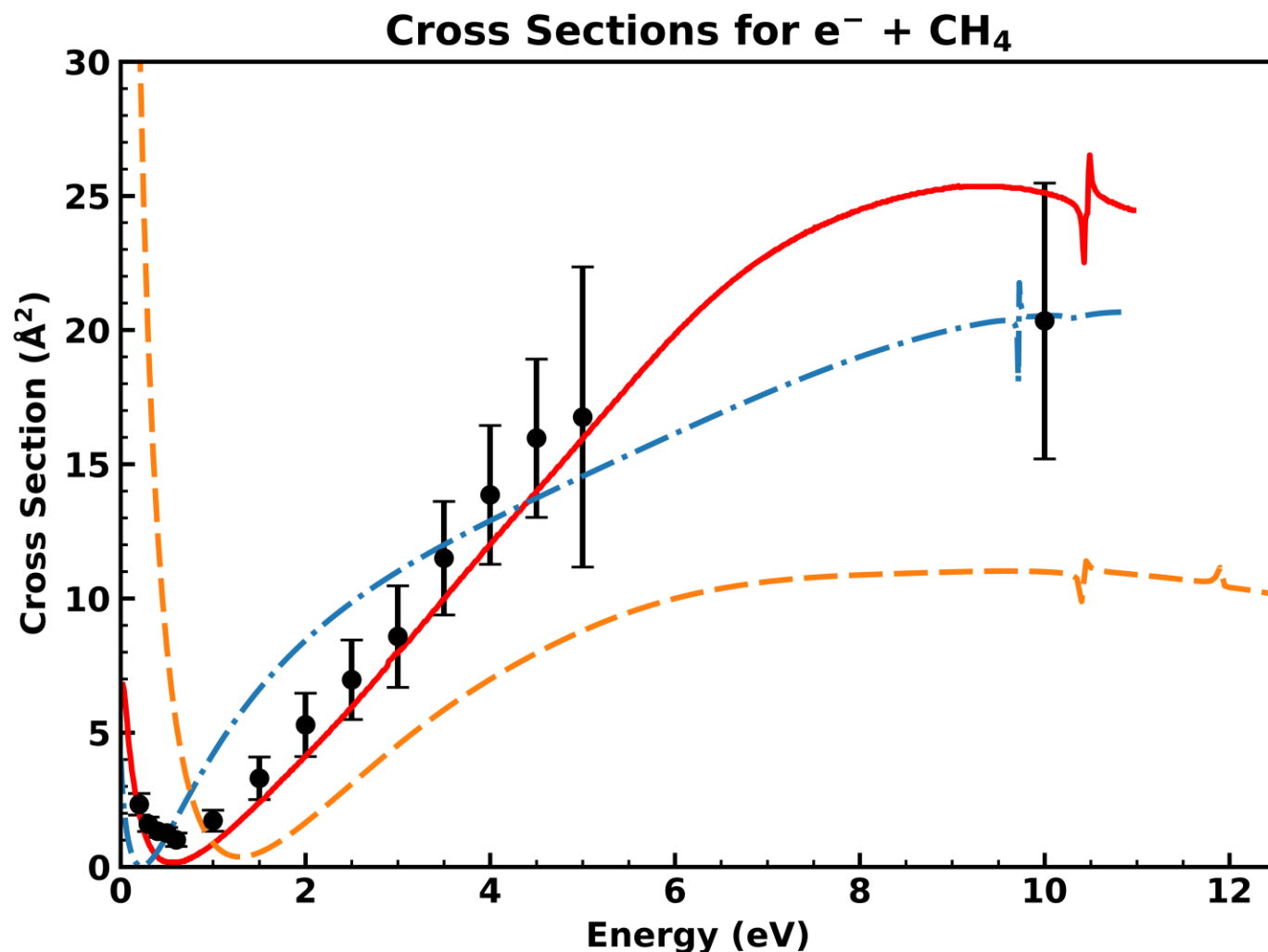
[ToF-MS]: Ferch *et al* *J. Phys. B.* **18**, L445 (1985).

[UKRMol]: Brigg *et al* *J. Phys. B.* **47**, 185203 (2014).

UKRMol+ / ASTRA Comparison

Literature Values:

- R-T minimum at 0.4 eV
- Resonance peak at 10.2 eV



	UKRMol+- CAS(8,8)- TS8-Lmax6	ASTRA- CAS(8,15)- Lmax4
Basis Set	cc-pVTZ	aug-cc-pVQZ
ΔE (eV)	11.341	9.776
R-T (eV)	1.301	0.232
Peak (eV)	10.451	9.727
RMSE (Å)	12.173	3.938

[UKRMol]: Brigg *et al* *J. Phys. B.* **47**,185203 (2014). [Recommended]: Song *et al* *J. Phys. Chem. Ref. Data* **44**, 023101 (2015).

Conclusions and Future Work

- Studied low-energy electron scattering calculations for CH₄ using UKRMol+ and ASTRA computational frameworks
- Both UKRMol+ and ASTRA captured R-T minimum and resonance peaks
 - Disagreements due to differences in how the target states are constructed, CC expansion, partial wave expansion, numerical methods
- Foundation for future investigations of more complex molecular targets
 - Tetrafluoromethane (CF₄), a PFAS compound used in plasma-based semiconductor manufacturing

Acknowledgments

- Dr. Siddhartha Chattopadhyay and Dr. Thomas C. Allison
- NIST SURF 2026 program

References

1. Brigg *et al J. Phys. B.* **47**, 185203 (2014).
2. Bundschu *et al J. Phys. B.* **30**, 2239 (1997).
3. Cho *et al J. Phys. B.* **41**, 045203 (2008).
4. Ferch *et al J. Phys. B.* **18**, L445 (1985).
5. Houfek *et al Comput. Phys. Commun.* **298**, 109113 (2024).
6. Lima *et al Phys. Rev. A.* **32**, 2696–2701 (1985).
7. Lohmann *et al J. Phys. B.* **19**, 2565 (1986).
8. Song *et al J. Phys. Chem. Ref. Data* **44**, 023101 (2015).
9. Tanaka *et al J. Phys. B.* **15**, 3305–3319 (1982).
10. Ziółkowski *et al J. Chem. Phys.* **137**, 22A510 (2012).